PATHS OF EVOLUTION

PYQ • 2021 - 2026

1. 2026 **2 MARKS**

(A) Structure of synapse is given. Analyse it and answer the questions.

(a) Identify the chemical labelled as 'X'.

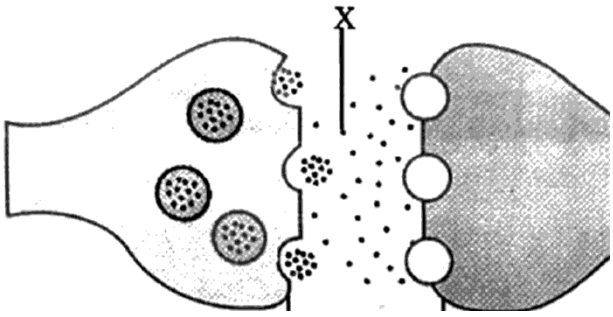
(b) Mention their role in the transmission of impulse to only one direction.

OR

(B) When we feel fear or tension, the heartbeat increases and digestion slows down without our control.

(a) Name the division of the autonomic nervous system that controls these actions.

(b) Write any two other actions controlled by this nervous system during such situations.



2. 2026 **3 MARKS**

Identify the fluid indicated by the statements given below and write any two of its functions.

• It fills the space between the inner layers of the meninges.

• Ependymal cells play a role in the formation of this fluid.

3. 2026 **3 MARKS**

Finches in Galapagos island : (A) Finches with beaks that are suitable for the available food (B) Finches with beaks that are not suitable for the available food

(a) What happens to finches in A and B categories in terms of survival ?

(b) What will happen to the population of finches that survive ?

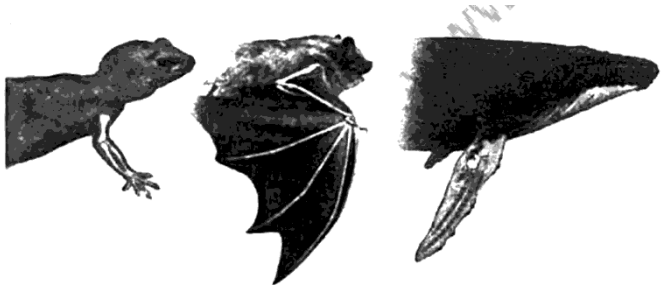
(c) How can these changes in survival lead to the formation of new species of finches ?

4. 2025 **2 MARKS**

Observe the given illustration and answer the questions.

(a) What similarity do these organs have in their internal structure ?

(b) How does the comparative study of these organs provide evidence for evolution ?

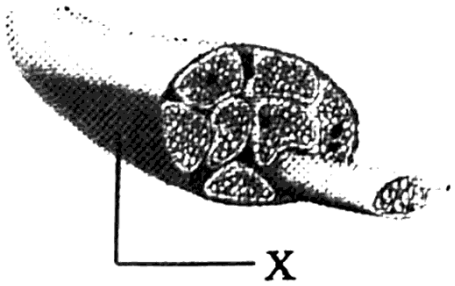


5. 2025 **2 MARKS**

Observe the cross-section of a nerve and write answers to the questions.

(a) What is the function of the part indicated by 'X' ?

(b) How do sensory nerves and motor nerves differ in function ?



6. 2025 **2 MARKS**

The main parts and functions of brain are given in the box. Arrange them in the following table suitably.

Thalamus, Controls voluntary movements, Cerebellum, Acts as relay station of impulses, Hypothalamus, Maintains equilibrium of the body, Cerebrum, Maintains homeostasis

7. 2025 **3 MARKS**

Observe the illustration regarding different types of synapses and answer the questions.

A : Neuron — Neuron B : Neuron — (i) C : Neuron — (ii)

(a) Identify and write the parts that are indicated by (i) and (ii).

(b) What is the role of neurotransmitters in the transmission of impulses across the synapse ?

(c) What are the two main functions of synapses ?

8. 2024 **1 MARK**

Which of the given statements is related to sympathetic system ?

• Part of autonomous nervous system.

• Part of central nervous system.

• Production of saliva increases.

• Glycogen is converted to glucose.

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9. 2024 **2 MARKS**

Observe the illustration and answer the questions.

Organisms with variations : Over production → Struggle for existence

(a) Which theory of evolution is indicated here ?

(b) What is the significance of variations in the origin of new species according to this theory ?

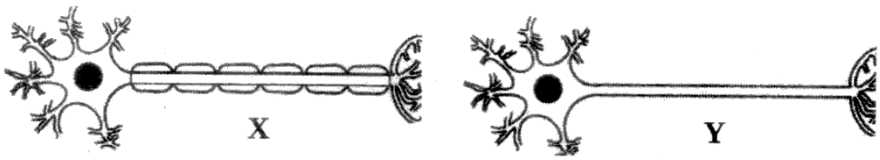
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10. 2024 **2 MARKS**

Observe the figures X, Y and answer the questions.

(a) Which neuron is present in abundance in the grey matter ? What features did you consider for the selection ?

(b) In which neuron is the speed of impulse high ? Why ?



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11. 2024 **3 MARKS**

Choose any one of the evidences given below and explain how it substantiates evolution.

(a) Palaeontology

(b) Physiology and Biochemistry

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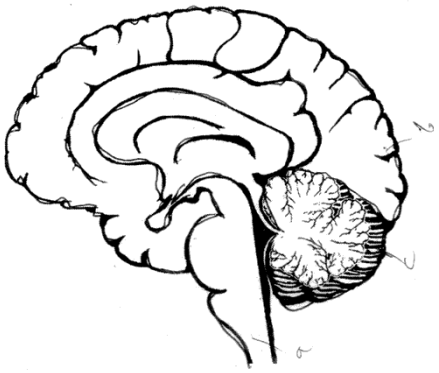
12. 2024 **4 MARKS**

Redraw the diagram. Identify and label the parts mentioned below.

(a) Part that controls involuntary actions.

(b) Part that evokes sensations.

(c) Part that receives impulses from the vestibular nerve.



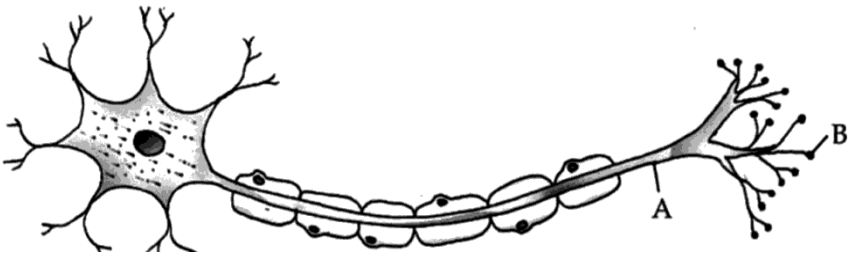
13. 2023 **1 MARK**

Find out the odd one and write the common feature of others.

Chimpanzee, Gorilla, Monkey, Orangutan

14. 2023 **2 MARKS**

Write the name and functions of A and B in the given figure.



15. 2023 **2 MARKS**

Analyse the given statements and answer the questions.

(i) Sudden withdrawal of the hand while touching on a hot object unknowingly.

(ii) Blinking of eyes when light suddenly falls on them.

(a) Name the reflex actions that take place in (i) and (ii).

(b) What is reflex action ?

16. 2023 **3 MARKS**

Arrange columns B and C according to column A.

A (Parts) : Cerebrum, Cerebellum, Medulla Oblongata

B (Peculiarity) : Rod shaped part; Situated just below the thalamus; The second largest part of the brain; The largest part of the brain

C (Function) : Maintenance of homeostasis; Controls involuntary actions like heart beat, breathing etc.; Evokes sensations; Maintains equilibrium of the body

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17. 2023 **3 MARKS**

The flow chart and hints related to a theory of evolution are given. Over production → (A) → [No favourable variations : Destroyed] / [Favourable variations → (B) → Survive]

Hints : Mutation, Natural selection, Struggle for existence, Acquired characters

(a) Choose suitable hints from the box and fill in A and B.

(b) Which theory is indicated in the flow chart ?

(c) What is the limitation of this theory ?

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18. 2022 **1 MARK**

Choose the action of sympathetic system from the following.

(a) Peristalsis become normal (b) Heart beat increases

(c) Gastric activities become normal (d) Trachea contracts

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19. 2022 **2 MARKS**

Observe the illustration and answer the questions.

Anthropoidea → (i) _____ and (ii) Hominoidea

(a) Fill (i).

(b) What are the characteristics of organisms belong to (ii) ?

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20. 2022 **4 MARKS**

The following are the main concepts of a theory of evolution. Analyse it and answer the questions.

(i) Accumulation of variations inherited through generations.

(ii) Origin of new species.

(iii) Struggle for existence.

(iv) Favourable variations are transferred to the next generation.

(v) Those with no favourable variations are destroyed and survival of others.

(vi) Over production.

(a) Name of the evolutionary theory.

(b) Rearrange the concepts given above in sequential order.

21. 2022 **5 MARKS**

Redraw the diagram, and answer the following questions.

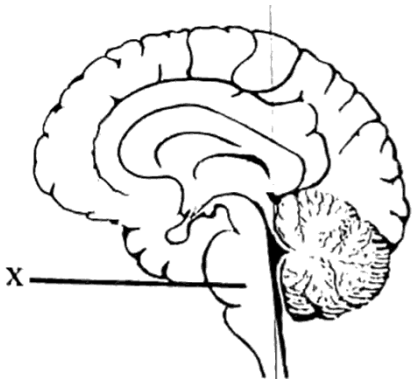
(a) Identify and label the parts with their names :

(i) Maintains equilibrium of the body

(ii) Controls voluntary movements

(iii) Acts as relay station of impulses

(b) Name the part labelled 'X' and write its function.



22. 2021 **1 MARK**

Identify the part of brain which helps to maintain homeostasis.

(a) Cerebrum (b) Thalamus (c) Medulla oblongata (d) Hypothalamus

23. 2021 **1 MARK**

Which organism belongs to cercopithecoidea ?

(a) Gorilla (b) Gibbon (c) Monkey (d) Chimpanzee

24. 2021 **2 MARKS**

We withdraw our hand when accidentally touch on a sharp object.

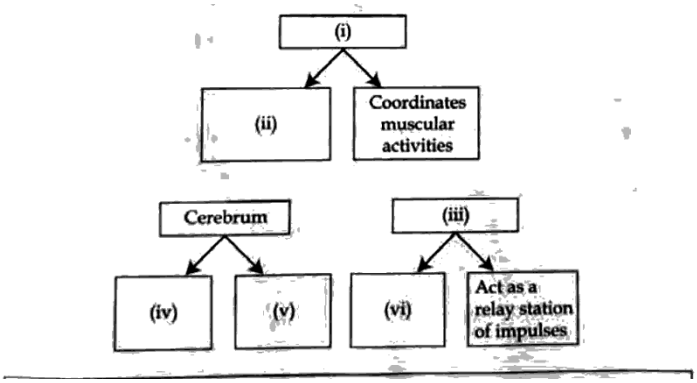
(a) What kind of response is mentioned here ?

(b) Name two types of such responses.

25. 2021 **3 MARKS**

Complete the following illustration using appropriate statements given in the box.

Box : Maintain equilibrium of the body ; Cerebellum ; Evoke sensation ; Centre of thought, intelligence, memory and imagination ; Thalamus ; Analyses impulses from various parts of the body and sends to the cerebrum



26. 2021 **3 MARKS**

Overproduction → (i) _____ → [No favourable variations : Destroyed] / [Favourable variations : Natural Selection]

(a) Fill up (i).

(b) How Natural Selection leads into the origin of new species ?

27. 2021 **4 MARKS**

Observe the figure and answer the questions.

(a) Identify the cell.

(b) Name the parts labelled A and B.

(c) Write down the function of C.

